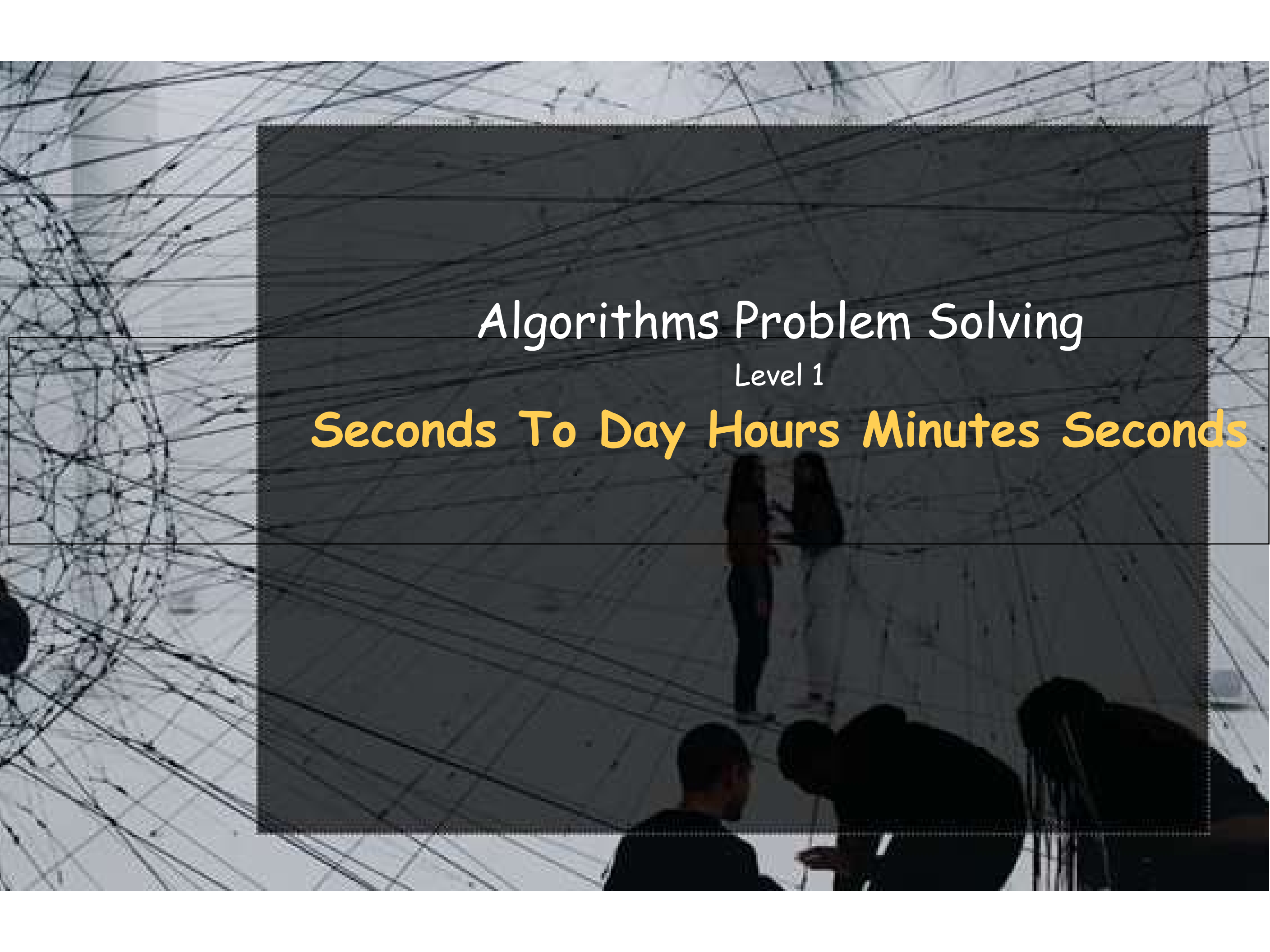


Algorithms Problem Solving

Level 1

Seconds To Day Hours
Minutes Seconds



Algorithms Problem Solving

Level 1

Seconds To Day Hours Minutes Seconds

Definition

- ◆ This program reads a total number of seconds and converts it into days, hours, minutes, and remaining seconds.
- ◆ The program uses a structure to store the converted time values.
- ◆ The conversion is done using integer division `/` and the remainder operator `%`.

Explanation

- ◆ ``strTaskDuration`` stores the four parts of the duration:
 - `NumberOfDays`
 - `NumberOfHours`
 - `NumberOfMinutes`
 - `NumberOfSeconds`
- ◆ ``ReadTotalSeconds()`` reads the total number of seconds from the user.
- ◆ ``TaskSecondsDuration()`` performs the conversion.
- ◆ ``SecondsPerDay`` is ``24 × 60 × 60 = 86400``.
- ◆ ``SecondsPerHour`` is ``60 × 60 = 3600``.
- ◆ ``SecondsPerMinute`` is ``60``.
- ◆ First, the total seconds are divided by the number of seconds in one day:
``TotalSeconds / SecondsPerDay``
- ◆ The remainder is stored because the seconds used for the complete days should no longer be considered.
- ◆ The remainder is then divided by the number of seconds in one hour to calculate the hours.
- ◆ ``%=`` updates the remainder by keeping only the seconds left after extracting the complete hours.
- ◆ The same process is repeated for minutes and seconds.

Steps in main()

1. Read the total number of seconds.
2. Calculate the number of complete days using `/`.
3. Calculate the remaining seconds using `%`.
4. Calculate the complete hours from the remainder.
5. Update the remainder using `%`.
6. Calculate the complete minutes.
7. Calculate the remaining seconds.
8. Store all values in the structure.
9. Print the converted duration.

💡 Example

◆ If the user enters:

193535 seconds

◆ Days:

$193535 / 86400 = 2$ days

◆ Remaining seconds:

$193535 \% 86400 = 20735$

◆ Hours:

$20735 / 3600 = 5$ hours

◆ Remaining seconds:

$20735 \% 3600 = 2735$

◆ Minutes:

$2735 / 60 = 45$ minutes

◆ Remaining seconds:

$2735 \% 60 = 35$ seconds

➡ Result

The result is 2 Days, 5 Hours, 45 Minutes and 35 Seconds.

Note

- ◆ Integer division `` / `` gives the number of complete units.
- ◆ The `` % `` operator gives the remaining part after extracting complete units.
- ◆ The conversion follows the same idea as #42, but in the opposite direction:
 - #42: Days, Hours, Minutes, Seconds → Total Seconds
 - #43: Total Seconds → Days, Hours, Minutes, Seconds

⊘ Warning

◆ The program does not validate whether the entered total seconds is positive. A negative value would produce results that do not represent a normal time duration.



Comparison



` / ` Integer Division

- ✓ Finds the number of complete units.
- ✓ Example: ``193535 / 86400 = 2``.
- ✓ Used to calculate days, hours, and minutes.



` % ` Remainder

- ✓ Finds what remains after complete units are extracted.
- ✓ Example: ``193535 % 86400 = 20735``.
- ✓ Used to continue converting the remaining seconds.

✦ Summary

- ✓ Total seconds can be converted into larger time units using integer division.
- ✓ ``%`` keeps the remaining seconds after extracting each unit.
- ✓ The conversion order is Days → Hours → Minutes → Seconds.
- ✓ A structure can store all parts of the converted duration.
- ✓ ``/`` finds complete units, while ``%`` finds the remainder.

Thank you 😊